

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0720

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:50

Annexure A

Scenario-Based

82/100

Code of Conduct:

I Qhasani Nikesh NS (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Qhasani Nikesh NS

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

ANSWER:

1. Smart Campus Network Subnet Allocation

Given Network Block: 192.168.100.0/24

A /24 network contains $2^{(32-24)} = 2^8 = 256$ total addresses (Range: 192.168.100.0 – 192.168.100.255).

Sequential allocation is performed for the three departments to avoid overlap.

A. School of Computing

- Host bits: $32 - 25 = 7$ bits
- Total Addresses: $2^7 = 128$ addresses
- Usable Host Addresses: $128 - 2 = 126$ usable hosts

- Network Address: 192.168.100.0
- Broadcast Address: 192.168.100.127
- Valid Host Range: 192.168.100.1 – 192.168.100.126

Parameter	IP Address / Value
Network Address	192.168.100.0
First Host Address	192.168.100.1
Last Host Address	192.168.100.126
Broadcast Address	192.168.100.127
Usable Hosts	126

B. School of Electronics

Next available address begins at .128.

- Host bits: $32 - 26 = 6$ bits
- Total Addresses: $2^6 = 64$ addresses
- Usable Host Addresses: $64 - 2 = \mathbf{62}$ usable hosts
- Network Address: 192.168.100.128
- Broadcast Address: 192.168.100.191
- Valid Host Range: 192.168.100.129 – 192.168.100.190

Parameter	IP Address / Value
Network Address	192.168.100.128
First Host Address	192.168.100.129
Last Host Address	192.168.100.190
Broadcast Address	192.168.100.191
Usable Hosts	62

C. School of Mechanical Engineering

Next available address begins at .192.

- Host bits: $32 - 27 = 5$ bits
- Total Addresses: $2^5 = 32$ addresses
- Usable Host Addresses: $32 - 2 = \mathbf{30}$ usable hosts
- Network Address: 192.168.100.192
- Broadcast Address: 192.168.100.223
- Valid Host Range: 192.168.100.193 – 192.168.100.222

Parameter	IP Address / Value
Network Address	192.168.100.192
First Host Address	192.168.100.193
Last Host Address	192.168.100.222
Broadcast Address	192.168.100.223

D. Remaining Unused Address Range

- **Unused Range:** 192.168.100.224 – 192.168.100.255
- Total Unused Addresses: $256 - 128 - 64 - 32 = 32$ addresses

Summary Allocation Table – Question 1

Department	CIDR	Network	Host Range	Broadcast	Usable
Computing	/25	192.168.100.0	.1 – .126	192.168.100.127	126
Electronics	/26	192.168.100.128	.129 – .190	192.168.100.191	62
Mechanical	/27	192.168.100.192	.193 – .222	192.168.100.223	30
Unused	—	192.168.100.224	.224 – .255	192.168.100.255	32 addresses

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

ANSWER:

2. Multinational Software Company Subnetting

Given Network Block: 10.10.0.0/16

A /16 prefix contains $2^{(32-16)} = 65,536$ total IP addresses (Range: 10.10.0.0 – 10.10.255.255).

Subnet Breakdown

A. Software Development (/24):

- Host bits: 8 | Total: 256 | Usable: 254
- Network: 10.10.0.0 | Broadcast: 10.10.0.255 | Range: 10.10.0.1 – 10.10.0.254

B. Quality Assurance (/25):

- Host bits: 7 | Total: 128 | Usable: 126
- Network: 10.10.1.0 | Broadcast: 10.10.1.127 | Range: 10.10.1.1 – 10.10.1.126

C. Human Resources (/26):

- Host bits: 6 | Total: 64 | Usable: 62
- Network: 10.10.1.128 | Broadcast: 10.10.1.191 | Range: 10.10.1.129 – 10.10.1.190

D. Executive Management (/27):

- Host bits: 5 | Total: 32 | Usable: 30
- Network: 10.10.1.192 | Broadcast: 10.10.1.223 | Range: 10.10.1.193 – 10.10.1.222

E. Remaining Unused Address Space:

- Range: 10.10.1.224 – 10.10.255.255 (Provides significant expansion space for future subnets).

Summary Allocation Table – Question 2

Department	CIDR	Network	Host Range	Broadcast	Usable
Software Dev	/24	10.10.0.0	10.10.0.1 – .254	10.10.0.255	254
QA	/25	10.10.1.0	10.10.1.1 – .126	10.10.1.127	126
HR	/26	10.10.1.128	10.10.1.129 – .190	10.10.1.191	62
Executive	/27	10.10.1.192	10.10.1.193 – .222	10.10.1.223	30
Unused	—	10.10.1.224	10.10.1.224 – 10.10.255.255	10.10.255.255	65,056

3.A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the /64 prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

ANSWER:

Smart Hospital IPv6 Network

Given IPv6 Address Block: 2001:0db8:abcd:0000:0000:0000:0000/64

IPv6 addresses are 128-bit numbers represented as eight 16-bit hexadecimal blocks.

A. Compressed Notation

1. Remove leading zeros: 2001:db8:abcd:0:0:0:0:0
2. Replace contiguous zero blocks with '::':

Result: 2001:db8:abcd::/64

B. Expanded Notation

Expanding '::' restores the five missing 16-bit zero groups:

Result: 2001:0db8:abcd:0000:0000:0000:0000/64

C. Network Prefix

The prefix length /64 indicates that the first 64 bits form the network component:

Network Prefix: 2001:db8:abcd:0000::/64

D. Number of Host Addresses

6. Remaining bits for Interface ID: $128 - 64 = 64$ bits.

Total host addresses: $2^{64} = 18,446,744,073,709,551,616$ addresses

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

ANSWER:

ISP Routing Table Analysis

Destination IP Address: 192.168.2.45

Routing Table Entries: 192.168.1.0/24, 192.168.2.0/24, 192.168.3.0/24

Analysis & Routing Decisions

- **Destination Subnet:** *The IP 192.168.2.45 falls in the range 192.168.2.0 – 192.168.2.255, corresponding to subnet **192.168.2.0/24**.*
- **Matching Entry:** The router matches the entry 192.168.2.0/24 in its table.
- **Next-Hop / Interface:** The table does not specify explicit next-hop IPs or interface names; the packet will be forwarded out of the physical/logical interface bound to 192.168.2.0/24.
- **Default Route:** If a destination IP does not match any specific route, the router uses the default route: **0.0.0.0/0**.

Parameter	Routing Value / Decision
Destination IP	192.168.2.45
Destination Subnet	192.168.2.0/24
Matching Route Entry	192.168.2.0/24
Forwarding Action	Outbound interface associated with 192.168.2.0/24
Default Route Fallback	0.0.0.0/0

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

ANSWER:

Administration and Finance LAN Setup

Subnet Mask /26: 255.255.255.192 (Block size: 64 addresses, Usable hosts per subnet: 62)

A. Administration LAN (192.168.10.0/26)

- Network Address: 192.168.10.0
- Broadcast Address: 192.168.10.63
- Valid Host Range: 192.168.10.1 – 192.168.10.62
- Default Gateway: 192.168.10.1

Administration LAN Computer Configuration:

Device Name	IP Address	Subnet Mask	Default Gateway
PC1	192.168.10.2	255.255.255.192	192.168.10.1
PC2	192.168.10.3	255.255.255.192	192.168.10.1
PC3	192.168.10.4	255.255.255.192	192.168.10.1

B. Finance LAN (192.168.10.64/26)

- Network Address: 192.168.10.64
- Broadcast Address: 192.168.10.127
- Valid Host Range: 192.168.10.65 – 192.168.10.126
- Default Gateway: 192.168.10.65

Finance LAN Computer Configuration:

Device Name	IP Address	Subnet Mask	Default Gateway
PC1	192.168.10.66	255.255.255.192	192.168.10.65
PC2	192.168.10.67	255.255.255.192	192.168.10.65
PC3	192.168.10.68	255.255.255.192	192.168.10.65

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context(2)	Shows good understanding with minor gaps(1.5)	Partial understanding; misses key aspects(1)	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved(2)	Identifies most key issues with minor omissions(1.5)	Identifies limited issues; some irrelevant points(1)	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario(2.5)	Applies concepts with minor errors(2)	Limited or partially correct application(1.5)	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale(2)	Makes reasonable decisions with some justification(1.5)	Makes weak decisions with limited justification(1)	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented(1.5)	Generally clear with minor issues(1)	Somewhat unclear or poorly structured(0.5)	Disorganized and difficult to understand

	Understanding of Scenario (20%)	Identification of Key Issues (20%)	Application of Concepts (25%)	Decision making (20%)	Clarity of Response (15%)	
Q1	1.5	2	2	1.5	1	8
Q2	1.5	1.5	2.5	1.5	1	8
Q3	1	2	2.5	2	1.5	9
Q4	2	2	2	2	1.5	9.5
Q5	2	1	1.5	1	1	6.5
						AI